MILITARY SPECIFICATION

MANUALS, TECHNICAL: TRANSPORTATION, HANDLING, INSPECTION, OPERATIONS, FIRING, POST FIRING AND DISSECTION PROCEDURES

(For Missile Motors)

This specification is approved for use within the Department of the Air Force , and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope: This specification describes the requirements for a manual to provide procedures for the static firing of missile motors at Hill Air Force Base (HAFB) facilities. The manual will include the procedures for transportation of missile motors from the HAFB storage facility/depot maintenance facility (DMF) to the test site, roll transfer into and out of the radiographic and other buildings, inspections required throughout the receipt to the test pad area, post firing inspections and dissection. The manual will also provide procedures for the static firing of the motor, calibration of instrumentation, connection of motor instrumentation to the land lines, data collection, handling, reduction and analysis, and the final report format.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

MILITARY SPECIFICATIONS

MIL-M-38784 Manuals, Technical: General Requirements for Preparation of

MIL-P-38790 Printing Production of Technical Manuals: General Requirements for

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

Air Force Regulations

AFR 127-100 Explosives Safety Standards

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The requirements of MIL-M-38784 apply.

3.2 Security classification. Every effort shall be made to include only unclassified information in the manual. In the event that classification is essential, security classification and other aspects shall be in accordance with MIL-M-38784.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC/LOLME, Wright-Patterson AFB, OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Identification of items. The official nomenclature and part number of items and special equipment shall be listed one time in the manual. Further identification shall be in short descriptive form, but definite enough to insure clear identification. The principal noun shall be consistent with those in Identification Lists (IL) and drawings. Part numbers or National Stock Numbers (NSN) shall be held to an absolute minimum and used only when no other means of positive identification will suffice.

3.4 Illustrations. Illustrations and diagrams shall be prepared in accordance with MIL-M-38784. Illustrations used shall be adequate to definitely supplement the text and contribute to its clarity to the extent that they are essential for enabling the user to obtain a clear understanding of the subject matter.

3.4.1 Illustration of procedures. Essential illustrations depicting mechanical operations shall be included in pertinent sections, as necessary. It is not necessary to illustrate each step of an overhaul procedures, such as the removal of screws with an ordinary screw driver, removing a cover plate after the screws have been removed, etc. Procedural illustrations should definitely supplement the text by clarifying procedures which are essential for reliability and safety or are not obvious.

3.5 Arrangement. The manual shall be prepared to the following arrangement.

TITLE PAGE
LIST OF EFFECTIVE PAGES
TABLE OF CONTENTS
LIST OF ILLUSTRATIONS
LIST OF TABLES
INTRODUCTION
SECTION I: GENERAL SAFETY REQUIREMENTS
SECTION II. HANDLING, TRANSPORTATION, AND INSPECTION
SECTION III. PREPARATION FOR FIRING
SECTION IV. PREFIRE FACILITY PREPARATION
SECTION V: TEST OPERATIONS AND MOTOR FIRING
SECTION VI: POST FIRE OPERATIONS

3.5.1 Title page and list of effective pages. The title page and list of effective pages shall be prepared in accordance with the requirements of MIL-M-38784 for standard size manuals.

3.5.2 Table of contents, list of illustrations, and list of tables. The requirements of MIL-M-38784 are applicable.

3.5.3 Introduction. The introduction shall contain an explanation of the purpose and scope of the manual and any other relevant information that will contribute directly to the use of the manual. The introduction shall explain that the instructions in the manual will outline in detail and sequence all operations and procedures required (using HAFB facilities and equipment), to be performed from the time the rocket motors are removed from and returned to the operational inventory (DMF), motor surveillance storage facilities (R&D) at HAFB, until handling, inspection and post firing inspections are complete (except static firing) and reduction. Also, that the instructions will be prepared in a manner to include two types of motors: (1) research and development surveillance; (2) operational inventory surveillance.

3.5.4 Section I: General Safety Requirements. This section shall contain the following statements: "The safety requirements and precautions set forth in AFR 127-100 will be complied with by munitions personnel during static firing opeations. All personnel engaged directly, as well as indirectly, in operations in which ammunitions, explosives and other hazardous material are involved should be thoroughly trained in explosive safety and capable of recognizing hazardous explosive exposures. Thinking safety and working safely must become a firmly established habit when working with or in the vicinity of items capable of exhibiting a hazard due to the nature of their explosive, flammable, or toxic fillers. The absence of a safety requirement in this manual or in AFR 127-100 does not necessarily indicate that no safeguards are needed. Safety precautions are included in procedures where it is required. If immediately dangerous munitions are encountered, all operations in the immediate vicinity will be shut down, personnel evacuated to a safe location and EOD or other authorized personnel called to render assistance in elimination of the hazard. Operations will not be resumed until the hazard has been eliminated."

3.5.5 Section II: Handling, Transportation and Inspection. This section shall provide sufficient information to remove the rocket motor from HAFB storage facilities, horizontal/vertical and DMF, and prepare and instrument rocket motor for road test, vibration test, and board course test. These instructions shall also cover preparation and loading instructions for transportation, including unloading from transporter and all inspections (including Non-Destructive Inspection) required to determine prefiring posture. The information shall be brief but factual, giving step by step procedures. Information covered by separate technical manuals, i.e., borescope, overhaul instructions, etc., shall not be included in this manual; merely reference made thereto.

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3.5.6 Section III: Preparation for Firing. This section shall provide procedures for the preparation and inspection of the motor for static firing, i.e., instrumentation; preparation and installation of data acquisition system (DAS), DAS checkout, and installation of firing harness and other related fixtures required for static firing of motor.

3.5.7 Section IV: Prefire Facility Preparation. This section shall provide procedures for test facility preparation (test pad, annex building, data recording building) test equipment verification, test pad instrumentation requirements, digital data reduction information and test data documentation. In addition, procedures for support systems preparation, rocket motor installation in test stand, and rocket motor alignment in test fixture shall be provided. All procedures necessary for setup, checkout and inspection of mechanical and electrical systems shall be provided.

3.5.8 Section V: Test Operations and Motor Firing. This section shall provide procedures for motor instrumentation, termination and calibration, thrust vector control system (TVCS) checkout, TVCS dry run performance, prefire operations motor firing. Emergency procedures for motor hangfire and motor malfunction are required.

3.5.9 Section VI: Post Fire Inspection. This section shall provide procedures for postfire inspections, calibrations, photography, test setup disassembly, test data review and documentation, test anomalies or motor malfunction, dissection of motor and components, and data reduction and analysis.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality. Review, validation, verification and final acceptance shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY

5.1 Delivery. Packaging, packing and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1 Intended use. Refer to 1.1.

6.2 Definitions. Abbreviations used herein and their meanings are:

DAS - Data acquisition system
DMF - Depot maintenance facility
EOD - Emergency ordnance disposal
NSN - National stock number
IL - Identification List
HAFB - Hill Air Force Base, Utah
R & D - Research and development
TVCS - Thrust vector control system

6.3 Data Requirements

When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph	Data requirement title	Applicable DID no.	Option
1-1	Technical, Manual: Transportation, Handling, Inspection, Operations, Firing, Post Firing, and Dissection Procedures	DI-M-3407	DI-M-6153

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Custodian:
Air Force - 16

Preparing Activity:
Air Force - 16

Reviewers:
Air Force - 01, 10, 99

(Project TMSS-F491)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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